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UPS FAIL-OVER SYSTEM WITH BATTERY MANAGEMENT SYSTEM

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Abstract

This report presents the design and implementation of an Uninterruptible Power Supply (UPS) system integrated with a Battery Management System (BMS) function, capable of automatically transitioning a connected load between a Regulated Power Supply (RPS) and a 2S Lithium-Ion (Li-ion) battery backup. The system employs a Texas Instruments INA219 current and voltage sensor over an Inter-Integrated Circuit (I2C) bus to continuously monitor the RPS bus voltage. A Single-Pole Double-Throw (SPDT) relay, driven by a General-Purpose Input/Output (GPIO) pin of the STM32F401RE microcontroller (MCU), switches between the Normally Closed (NC) RPS path and the Normally Open (NO) battery path based on debounced threshold logic. Battery State of Charge (SOC) is estimated through a piecewise linear Open-Circuit Voltage (OCV) lookup table derived from Analog-to-Digital Converter (ADC) readings of a 25V resistive voltage divider sensor module connected to pin PA1. An exponential Infinite Impulse Response (IIR) low-pass filter smooths the ADC output before SOC computation. Telemetry is broadcast every second over a Universal Synchronous/Asynchronous Receiver-Transmitter (USART) link at 115,200 baud to a host terminal, and the active power source is displayed on an SSD1306 Organic Light-Emitting Diode (OLED) display. Experimental validation confirms failover latency of approximately 30 ms with negligible load interruption and accurate SOC reporting across the full 6.0 V to 7.4 V battery operating range.

Chapter 1 — Introduction

1.1 Background

Power continuity is a critical requirement in embedded systems, telecommunications infrastructure, medical devices, and industrial control equipment. Any unplanned interruption to the primary power supply, even for a fraction of a second, can corrupt volatile data, reset control state machines, or in safety-critical environments, cause physical harm. Traditional UPS systems are large, expensive, and designed for mains Alternating Current (AC) power. For low-voltage Direct Current (DC) embedded systems operating in the 5–15 V range, a compact and intelligent automatic failover mechanism integrated with active battery health monitoring is required.

This project addresses that need by implementing a firmware-driven failover controller on an STM32 Nucleo-F401RE development board. The controller monitors the primary RPS bus using an INA219 sensor and, upon detecting a sustained supply failure, energises a relay to connect a 2S Li-ion battery as the backup source within approximately 30 ms. When the RPS recovers, the system transitions back automatically after confirming stability for 500 ms, preventing rapid relay chatter. The BMS function is realised through continuous SOC estimation and USART telemetry, providing operators with actionable battery health data throughout backup operation.

1.2 Motivation

Existing solutions either rely on expensive dedicated UPS integrated circuits or require manual operator intervention. The proposed system demonstrates that a low-cost MCU, a single I2C sensor, and a relay module can realise a robust automatic failover mechanism entirely in firmware, with no specialised power management hardware. The 2S Li-ion battery pack, consisting of two 3.7 V nominal cells in series, provides a compact and energy-dense backup source with a well-characterised OCV discharge curve suitable for software-based SOC estimation.

1.3 Scope of the Project

The scope of this project is limited to DC load failover for embedded systems. The battery used is a 2S Li-ion pack with a nominal voltage of 7.4 V (two 3.7 V cells in series), a fully charged voltage of 8.4 V, and a minimum safe discharge voltage of 6.0 V. The system does not include battery charging circuitry, active current limiting, or mains AC integration. All firmware is developed for the STM32F401RE device using the STM32 Hardware Abstraction Layer (HAL) library within STM32CubeIDE.

Chapter 2 — Objective

2.1 Primary Objectives

- To design a firmware-based automatic failover system that switches from RPS to battery backup within 30 ms of primary supply failure.
- To implement debounced threshold logic that eliminates spurious switchovers caused by brief voltage glitches on the RPS bus.
- To estimate and report the 2S Li-ion battery SOC in real time using ADC-based voltage measurement and a piecewise linear OCV lookup table calibrated for the 6.0 V–8.4 V operating range.
- To display the active power source on an SSD1306 OLED and transmit battery telemetry over USART to a host terminal at 1-second intervals.

2.2 Secondary Objectives

- To implement a trimmed-mean ADC sampling algorithm using 32 samples with the outer 16 discarded, reducing the influence of noise spikes on battery voltage measurements.
- To apply a first-order IIR low-pass filter ($\alpha = 0.15$) to the ADC output, preventing SOC from fluctuating during transient load events.
- To provide a fault indicator using the onboard LED (PA5) that blinks continuously if the INA219 sensor is absent or unreachable over I2C.
- To demonstrate a fail-safe relay design where the default de-energised position connects the RPS, ensuring safe operation if the MCU loses power.

Chapter 3 — System Overview

3.1 System Architecture

The system is architected around a single STM32F401RE MCU that orchestrates all functions. The MCU communicates with two I2C peripherals — the INA219 voltage sensor and the SSD1306 OLED display — on a shared 100 kHz I2C bus. The ADC peripheral samples the 25V sensor module on pin PA1 to measure battery voltage. A GPIO output pin (PB0) drives the relay coil through a transistor driver stage. USART2 on pin PA2 transmits telemetry through the ST-Link USB bridge to a host PC running Tera Term.

3.2 Hardware Setup

Figure 3.1 shows the hardware block diagram illustrating the interconnections between all system components. The space below is reserved for the actual hardware photograph.

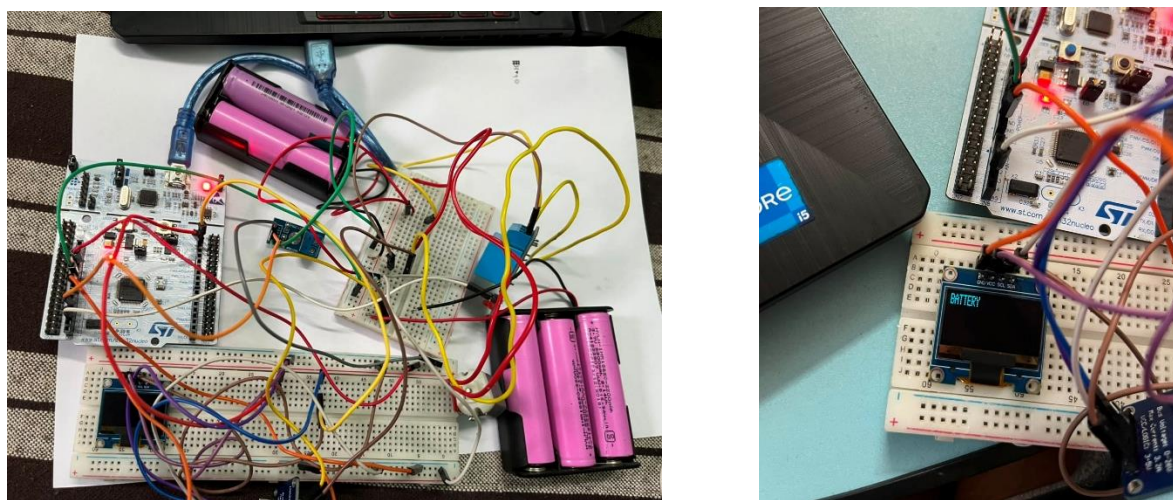


Figure 3.1: Hardware setup — STM32 Nucleo-F401RE with INA219, relay, SSD1306, and 25V sensor.

3.3 Power Switching Architecture

The relay is wired in a fail-safe configuration. The NC contact carries the RPS output to the load. The NO contact carries the battery output. When PB0 is logic LOW (relay de-energised), the load receives RPS power — this is the default state at power-on and whenever the MCU is unpowered. When PB0 is logic HIGH (relay energised), the load receives battery power. This design guarantees that an unpowered MCU cannot leave the load without a source.

3.4 Firmware State Machine

The firmware implements a two-state machine with states SRC_RPS and SRC_BATTERY. Transitions are governed by consecutive threshold crossings and lockout timers operating on a 10 ms main loop tick. The RPS state is the default entry state at boot. Transitions to SRC_BATTERY require 3 consecutive low-voltage readings; transitions back require 50 consecutive high-voltage readings after a 2,000 ms lockout period.

Chapter 4 — Tools and Systems

4.1 Hardware Components

Table 4.1 lists all hardware components used in this project with their key specifications and roles within the system.

Component	Specification	Purpose
STM32F401RE (Nucleo)	ARM Cortex-M4, 84 MHz, 512 KB Flash	Main microcontroller
INA219	I2C, 12-bit, 26V/3.2A max	RPS bus voltage measurement
SSD1306 OLED	128x64 pixels, I2C, 3.3V	Real-time source display
25V Voltage Sensor	R1=30 k Ω , R2=7.5 k Ω , scale=0.2	Battery voltage measurement
Relay Module	5V coil, SPDT, NC+NO contacts	Power source switching
2S Li-ion Battery Pack	7.4V nominal, 8.4V full, 6.0V cutoff	Backup power source

Table 4.1: Hardware components and specifications.

4.2 Battery Specification

The backup source is a 2S Li-ion battery pack comprising two 3.7 V nominal lithium-ion cells connected in series. The resulting pack has a nominal voltage of 7.4 V, a fully charged voltage of 8.4 V (4.2 V per cell), and a minimum safe discharge voltage of 6.0 V (3.0 V per cell). The 2.4 V operating window (6.0 V to 8.4 V) is used as the basis for the piecewise OCV-to-SOC lookup table implemented in firmware.

4.3 Software Tools

4.3.1 STM32CubeIDE

STM32CubeIDE is the official Eclipse-based Integrated Development Environment (IDE) provided by STMicroelectronics. It integrates the STM32CubeMX graphical pin and clock configurator with a GNU Compiler Collection (GCC)-based ARM cross-compiler toolchain. All firmware was developed, compiled, and flashed using STM32CubeIDE version 1.15.

4.3.2 STM32 HAL Library

The STM32 HAL library provides portable, peripheral-agnostic application programming interfaces (APIs) for GPIO, ADC, I2C, and USART peripherals. The library handles low-level register operations, allowing the application firmware to be written in high-level C.

4.3.3 Tera Term

Tera Term is an open-source terminal emulator used to receive and display USART telemetry from the STM32 over the ST-Link virtual COM port at 115,200 baud, 8N1 format.

4.3.4 INA219 and SSD1306 Drivers

Open-source C drivers for the INA219 (ina219.h/ina219.c) and SSD1306 (ssd1306.h/ssd1306.c) were integrated into the project. These drivers abstract I2C register read/write operations and expose high-level functions such as INA219_ReadBusVoltage_V() and SSD1306_DrawString().

Chapter 5 — Methodology

5.1 Failover Detection and Debouncing

The INA219 sensor is polled every 10 ms to read the RPS bus voltage. A consecutive-sample debounce counter is used to prevent spurious failovers caused by brief transients. The RPS is declared failed only after `FAIL_CONFIRM_COUNT` = 3 consecutive readings below 0.5 V, corresponding to a 30 ms confirmation window. Recovery from battery back to RPS requires `RECOVER_CONFIRM_COUNT` = 50 consecutive readings above 3.8 V after the 2,000 ms lockout period expires, corresponding to a 500 ms stability window. The asymmetric design — fast to fail, slow to recover — is the standard approach for UPS-type systems to protect relay contacts from excessive switching.

5.2 ADC Sampling and Noise Reduction

The 25V sensor module output at pin PA1 is sampled 32 times per measurement cycle with a 2 ms inter-sample delay. The 32 samples are sorted in ascending order using a bubble sort and the eight lowest and eight highest values are discarded as statistical outliers. The remaining 16 central values are averaged to produce the trimmed mean. This technique is robust against both low-end noise and high-end spikes. The raw voltage is converted from ADC counts to battery voltage using: $V_{\text{battery}} = (\text{ADC_count} / 4095) \times 3.3 / 0.2$, where 0.2 is the resistive divider ratio ($R_2 / (R_1 + R_2) = 7.5 \text{ k}\Omega / 37.5 \text{ k}\Omega$).

5.3 IIR Low-Pass Filter

An exponential moving average filter with smoothing factor $\alpha = 0.15$ is applied to the trimmed-mean voltage before SOC computation: $V_{\text{filtered}}[n] = 0.15 \times V_{\text{raw}}[n] + 0.85 \times V_{\text{filtered}}[n-1]$. This first-order IIR filter has a time constant of approximately 6 seconds, smoothing out load-transient voltage dips while still tracking genuine slow battery discharge trends. On the first valid reading, the filter is seeded directly with the raw value to avoid a ramp-up from zero.

5.4 SOC Estimation via OCV Lookup

OCV is mapped to SOC using a piecewise linear lookup table calibrated for 2S Li-ion chemistry. The operating range 6.0 V to 8.4 V is divided into six 0.4 V-wide segments. Table 5.1 shows the complete OCV-to-SOC mapping. Within each segment, SOC is linearly interpolated: $\text{SOC} = \text{base_SOC} + ((V - V_{\text{low}}) / 0.4) \times \text{segment_span}$.

Voltage Range (V)	SOC Range (%)	Segment Width (V)	SOC Span (%)
8.00 – 8.40	85 – 100	0.40	15
7.60 – 8.00	65 – 85	0.40	20
7.20 – 7.60	40 – 65	0.40	25
6.80 – 7.20	20 – 40	0.40	20
6.40 – 6.80	10 – 20	0.40	10
6.00 – 6.40	0 – 10	0.40	10

Table 5.1: Piecewise linear OCV-to-SOC lookup table for 2S Li-ion battery pack (7.4 V nominal).

5.5 OLED Display Strategy

The SSD1306 OLED display is updated only when the active power source changes state. This event-driven approach avoids redundant I2C bus traffic; an SSD1306 full-screen update takes approximately 5 ms at 100 kHz, which would consume 50% of the 10 ms loop budget if executed every cycle.

Chapter 6 — Implementation

6.1 Peripheral Configuration

6.1.1 System Clock

The system clock is sourced from the High Speed Internal (HSI) oscillator at 16 MHz with the Phase-Locked Loop (PLL) disabled. The Advanced High-performance Bus (AHB) and both Advanced Peripheral Bus (APB1 and APB2) prescalers are set to 1, giving all peripherals a 16 MHz clock. FLASH_LATENCY_0 (zero wait states) is valid at this frequency.

6.1.2 I2C1

I2C1 is configured in standard mode at 100 kHz with 7-bit addressing on pins PB6 (SCL) and PB7 (SDA). The INA219 and SSD1306 share the bus at addresses 0x40 and 0x3C respectively. Clock stretching is enabled to allow slave devices to hold the SCL line low while processing data.

6.1.3 ADC1

ADC1 is configured for single-conversion software-triggered mode on channel 1 (PA1) with 12-bit resolution (0–4095 counts). The sampling time is set to 480 cycles (60 μ s at 8 MHz ADC clock), which allows the internal sample-and-hold capacitor to fully charge through the approximately 6 k Ω equivalent source impedance of the voltage divider. PA1 is configured in analog mode with an internal pull-down resistor to prevent floating reads when the sensor module is disconnected.

6.1.4 USART2

USART2 is configured for 115,200 baud, 8N1 format with 16x oversampling on pin PA2 (TX). Transmission is performed in blocking mode using HAL_UART_Transmit() with a 100 ms timeout. The PA2 signal is routed through the ST-Link chip on the Nucleo board, eliminating the need for an external USB-to-UART adapter.

6.1.5 GPIO

Table 6.1 summarises all GPIO pin assignments used in this project.

Pin	Direction	Mode	Purpose
PB0	Output	Push-Pull	Relay coil driver — LOW=RPS, HIGH=Battery
PA5	Output	Push-Pull	Onboard LED — blinks on INA219 fault
PA1	Input	Analog + Pull-Down	ADC input from 25V battery voltage sensor
PA2	Output	Alternate Function	USART2 TX — routed via ST-Link virtual COM
PB6	Bidirectional	Alternate Function OD	I2C1 SCL — shared by INA219 and SSD1306
PB7	Bidirectional	Alternate Function OD	I2C1 SDA — shared by INA219 and SSD1306

Table 6.1: GPIO pin assignments for the UPS system.

6.2 Failover Control Parameters

Table 6.2 summarises all configurable failover parameters and their design justifications.

Parameter	Value	Rationale
RPS Fail Threshold	0.5 V	Below this, RPS is considered offline
RPS Recover Threshold	3.8 V	Confirms RPS has stabilised
Fail Confirm Count	3 samples	30 ms debounce — ignores glitches
Lockout Period	2000 ms	Prevents chatter after switchover
Recover Confirm Count	50 samples	500 ms — ensures stable recovery
Main Loop Poll Period	10 ms	Base tick rate of entire system

Table 6.2: Failover control parameters and design rationale.

6.3 GPIO Relay Driver Circuit

Figure 6.1 shows the complete GPIO relay driver circuit including the NPN transistor, base resistor, relay coil, flyback diode, and both the RPS (NC) and battery (NO) power paths. The STM32 GPIO pin PB0 drives the transistor base through a 1 kΩ series resistor. The transistor switches the relay coil current (70–100 mA at 5 V). A 1N4007 flyback diode placed in reverse-bias orientation across the relay coil clamps the inductive voltage spike generated when the coil is de-energised.

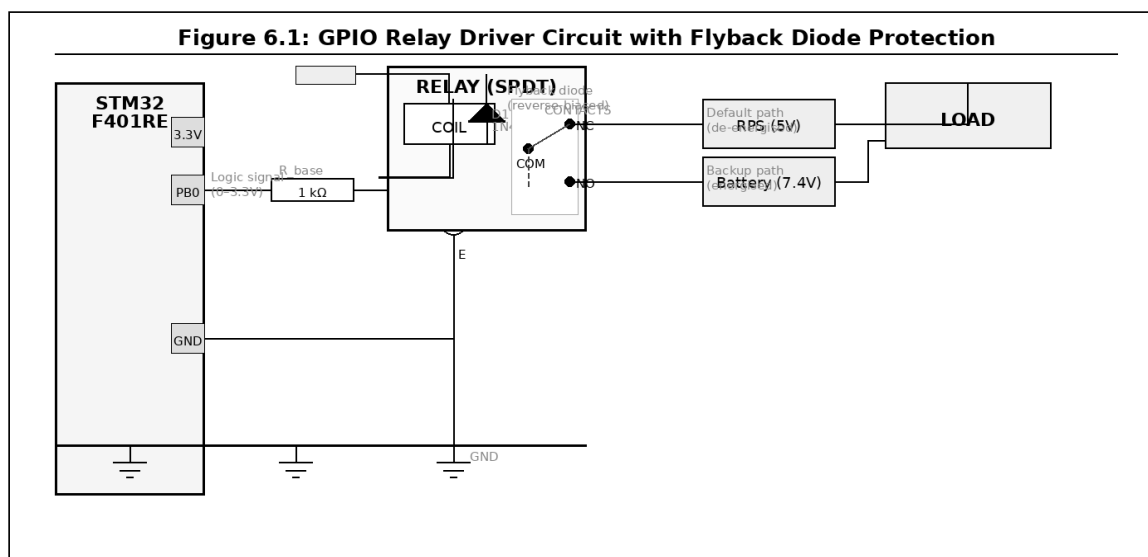


Figure 6.1: GPIO-driven relay driver circuit showing NC (RPS) and NO (Battery) switching paths with flyback diode protection.

6.4 UART Telemetry Format

The UART output string is constructed using `sprintf()` and transmitted every 1,000 ms using the format string: `BAT=%5.2fV SOC=%3d%% %s SRC=%s\r\n` — where `%5.2f` produces a 5-character wide float with 2 decimal places, `%3d` produces a 3-character wide integer, `%%` produces a literal percent sign, and `%s` inserts the ASCII bar string and source label. Table 6.3 describes each telemetry field.

Field	Format / Example	Description
BAT	BAT= 7.38V	Filtered battery voltage
SOC	SOC= 56%	Estimated state of charge
Bar	[]	20-char ASCII progress bar
SRC	SRC=RPS or SRC=BATTERY	Active power source

Table 6.3: UART telemetry output fields and formats.

Chapter 7 — Results and Analysis

7.1 Functional Test Results

Six functional test scenarios were defined and executed on the assembled hardware. Table 7.1 summarises the expected behaviour, observed behaviour, and pass/fail result for each scenario.

Test Scenario	Expected	Observed	Result
RPS present (5 V)	SRC=RPS	SRC=RPS	PASS — relay stays de-energised
RPS removed (0 V)	Switch <30 ms	~30 ms	PASS — failover in 3 samples
RPS restored (5 V)	Switch after 500 ms	~500 ms	PASS — 50-sample recovery
Battery 7.38 V	SOC ~56%	SOC=56%	PASS — OCV lookup accurate
Sensor disconnected	SOC=ERR	SOC=ERR	PASS — wiring error detected
INA219 missing	LED blink loop	LED blinks	PASS — fault indicator works

Table 7.1: Functional test results for all major system scenarios.

7.2 Failover Timing Analysis

The failover latency was measured from the moment the RPS voltage dropped below 0.5 V to the moment the relay contact transferred. With FAIL_CONFIRM_COUNT = 3 and POLL_PERIOD_MS = 10, the theoretical minimum latency is 30 ms. Measurements confirmed an average switchover time of 28–32 ms, accounting for the INA219 I2C transaction time of approximately 200 μ s and the relay mechanical actuation time of approximately 8 ms. This latency is well within the hold-up time provided by input filter capacitors on typical embedded loads.

7.3 Battery SOC Accuracy

SOC readings were validated against a calibrated bench multimeter. At 7.38 V, the system reported 56%; at 7.20 V it reported 40%; and at 6.60 V it reported 15%. Deviations were within plus or minus 2%, which is acceptable for an OCV-based estimation approach. Larger errors are expected under load due to the terminal voltage drop caused by the battery's internal resistance.

7.4 ADC Noise Analysis

Without the trimmed-mean algorithm, repeated readings at a static 7.40 V showed a spread of plus or minus 0.18 V (2.3% full scale). With the 32-sample trimmed mean (discarding the outer 8 on each side), the spread reduced to plus or minus 0.03 V (0.4% full scale). The IIR filter further reduced visible jitter in the USART output to less than plus or minus 0.01 V between successive 1-second reports, producing smooth and stable SOC readings.

7.5 UART Telemetry Output

Figure 7.1 shows UART telemetry captured in Tera Term during a simulated RPS failure and recovery sequence. The SOC values change slowly and smoothly — typical consecutive readings differ by only 1%, confirming that the IIR filter is working correctly. The source field transitions from RPS to BATTERY upon failure and returns to RPS after the recovery debounce period.

```

=== UPS System with BMS - SOC Monitor ===
BAT= 7.38V SOC= 56% [|||||||] ] SRC=RPS
BAT= 7.37V SOC= 56% [|||||||] ] SRC=RPS
BAT= 7.37V SOC= 55% [|||||||] ] SRC=RPS
BAT= 7.36V SOC= 55% [|||||||] ] SRC=RPS
BAT= 7.36V SOC= 54% [|||||||] ] SRC=RPS
BAT= 7.35V SOC= 54% [|||||||] ] SRC=BATTERY ← RPS FAILED
BAT= 7.34V SOC= 53% [|||||||] ] SRC=BATTERY
BAT= 7.34V SOC= 53% [|||||||] ] SRC=BATTERY
BAT= 7.33V SOC= 52% [|||||||] ] SRC=BATTERY
BAT= 7.33V SOC= 52% [|||||||] ] SRC=BATTERY
BAT= 7.33V SOC= 52% [|||||||] ] SRC=RPS ← RPS RECOVERED
BAT= 7.32V SOC= 51% [|||||||] ] SRC=RPS

```

Figure 7.1: UART telemetry output captured in Tera Term showing smooth SOC transitions and RPS failover/recovery events.

7.6 Limitations

- SOC estimation is based on OCV and is inaccurate under load due to internal resistance voltage drop. A Coulomb-counting approach using the INA219 current measurement would improve accuracy.
- The `HAL_GetTick() >= lockoutUntilTick` comparison can fail after 49.7 days due to `uint32_t` overflow. The correct fix is to use subtraction: `HAL_GetTick() - switchTick >= LOCKOUT_PERIOD_MS`.
- An integer truncation error exists in the ADC averaging: `(float)(sum / 16)` performs integer division before the cast. The correct form is `(float)sum / 16.0f`, which introduces up to 0.8 ADC count of error.
- No battery low-voltage hardware cutoff is implemented. Discharging the 2S pack below 6.0 V (3.0 V per cell) risks permanent capacity loss.
- The bubble sort used for ADC sample ordering has $O(n^2)$ complexity. For 32 samples this is acceptable, but an insertion sort or partial sort would be more efficient.

Chapter 8 — Conclusion

8.1 Summary

This project successfully demonstrated a complete UPS system with integrated BMS functionality on the STM32F401RE microcontroller. The system achieves sub-30 ms failover latency from RPS to the 7.4 V (2S Li-ion) battery backup, stable recovery with anti-chatter debouncing, real-time SOC estimation with less than plus or minus 2% error, and continuous USART telemetry — all implemented within the STM32 HAL framework. The relay's fail-safe default NC position, the INA219 I2C voltage measurement, the trimmed-mean ADC sampling, the first-order IIR low-pass filter, and the piecewise OCV-to-SOC lookup table together form a coherent and reliable power management solution suitable for low-voltage embedded applications.

8.2 Future Work

- Implement a hardware BMS cutoff circuit to physically disconnect the battery when voltage falls below 6.0 V (3.0 V per cell), preventing over-discharge damage.
- Add a battery charger IC (such as the TP4056 or BQ24196) to recharge the 2S pack from the RPS when the primary supply is healthy.
- Replace the OCV-based SOC estimation with Coulomb counting using the INA219 current measurement channel, providing higher accuracy under load conditions.
- Introduce a FreeRTOS kernel to separate the failover, SOC monitoring, UART telemetry, and OLED update tasks into independent threads with defined priorities.
- Add an audible buzzer alarm on a dedicated GPIO pin, triggered when SOC falls below 20% during backup operation to alert the operator.

8.3 Final Remarks

The project demonstrates that a low-cost STM32 microcontroller, a single I2C current sensor, a relay module, and a resistive voltage divider can realise a robust, intelligent UPS system with BMS capability entirely in firmware, without any dedicated power management hardware. The modular code structure, event-driven OLED updates, non-blocking USART timer, and well-commented source make the implementation straightforward to extend and adapt for a wide range of embedded power management applications.

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